PA194C and PA194CS: two different 194-state PA consistency machines

A complete corpus comparison, exact construction, and attempts to cross the 194-state frontier

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Abstract

We deliver two complete binary, single-tape, blank-start machines with **194 working states**, called PA194C and PA194CS. The requested strict reduction below 194 was not obtained. Both fallbacks are materially different from the three positive 194-state constructions in the supplied corpus. PA194C returns successful modus-ponens conclusions in a cache and tests that cache in Shift. PA194CS further removes a restoration loop by using symmetry of a destructive equality routine, and recovers its state cost with a new contextual fold. Their sources have 3,649 and 3,646 primitive instructions, compared with 3,673 in the supplied constructions. Both control graphs are new even after state renaming, change of designated start, tape reflection and symbol complementation, and are distinct from one another under those transformations.

Both machines retain the 67-reader/seven-fold axiom decoder. PA194CS adds one controller reader and folds it with an existing decrement interface. Both verification implementations pass all 4,749 and 4,745 respective arbitrary-register instruction outcomes. The independent literal-tape checker rederives its list lemmas from the actual rows and checks complete blank initializations of 57,922 and 57,768 transitions. Ordinary finite-trace preservation connects both machines to the supplied 195C source. Composing this with the separate-head calculus's ordinary syntactic PA calibration gives $\mathrm{PA} \vdash (\neg H_M \leftrightarrow \mathrm{Con(PA)})$ for each delivered machine. The logical calibration is an explicitly inherited mathematical dependency; no new end-to-end proof-assistant theorem is claimed.

The comparison is pinned to ProveIt commit `768c2be1a8e4d2fcf8aa3443c7b0f084d145734c`. All 36 incoming archives were traversed, including nested packages, and direct PA tables were included. We distinguish supported manuscript constructions, conditional formal endpoints and rejected experiments. The archive includes both complete transition tables, deterministic builders, two verification implementations, source maps, evidence, search records, the corpus census and this report's LaTeX source.

Contents

1 Result, conventions and exact artifact **3**
1.1 What is new, and what is inherited . . . 3

2 Comparison of the entire supplied corpus **4**
2.1 Snapshot and census scope . . . 4
2.2 All existing 194-state constructions . . . 5
2.3 The complete 195-state frontier . . . 5
2.4 How the larger designs relate . . . 6

3 A different 194-state controller: PA194C **6**
3.1 The exact transfer theorem . . . 7
3.2 Cached-head commands and their completeness . . . 8
3.3 The inherited logical calibration and proof boundary . . . 9

4 A second 194-state machine from a symmetric guard **10**
4.1 The comparison routine is symmetric in its full effect . . . 10
4.2 The shortened Shift . . . 11
4.3 Recovering the state cost by an exact contextual fold . . . 12
4.4 Consistency calibration and verification scope . . . 12

5 Why the symbolic physical checks are universal **13**
5.1 Verification results . . . 14

6 Logical calibration and its exact dependency **14**

7 Search for fewer than 194 controls **15**
7.1 Decoder allocation and contextual quotients . . . 15
7.2 Whole-controller screening . . . 16
7.3 Further source-level work . . . 17

8 Ideas transferred from ZFC and ConciseIPRA **17**

9 Reproduction and package contents **18**

A Complete PA194C transition table **19**

B Complete PA194CS transition table **20**

C Independent proof of the symmetric destructive guard **21**

D Every incoming archive **23**

1 Result, conventions and exact artifact

The first artifact is `construction/candidate/pa194C.tm`. Its working controls are $0, \ldots, 193$, its start is 0, and its only terminal control is 194. The tape alphabet is $\{0, 1\}$, the tape is two-way infinite and initially all zero, and the head initially scans cell zero. Each transition writes one bit and moves one cell left or right. There are 388 rows and no outgoing row for the terminal control. All 194 working controls are reachable in the pointed transition graph. This is graph reachability, not a claim that the blank computation visits every control.

The dense file's SHA-256 is

`d759313c03b515293fe06b12769fa1771d3829a2bad89febaa9b4391990f3a4b`.

Its native presentation, with start 74 and halt 254, has SHA-256

`b35d80b01ce9c2e95c8b29222881f40928247cdeffac322e4643f4f6694cd33c`.

The supplied label list is a checked conjugacy of all 388 rows. Appendix A prints the complete dense table. The second artifact is `secondary/pa194CS/machine/pa194CS.tm`, with the same state, start and halt conventions; its definition and hash are given in Section 4, and its complete table in Appendix B.

1.1 What is new, and what is inherited

The first contribution is a new verified composition at the current state frontier. Neither the cached-head command convention nor the seven-fold decoder is claimed as a newly invented ingredient. Both were supplied, but they had not been combined into a 194-state machine in the pinned corpus. The existing 194-state machines use the original Shift/MP controller. PA194C changes six native transition rows, two source-tree definitions, the full source graph, and command-level traces. It retains the exact source of the supplied 195C machine while reducing that machine's physical allocation by one control.

The second contribution is a further source-level simplification: PA194CS exploits the fact that the existing equality test clears both arguments on failure. It can exchange those arguments and omit an entire restoration loop. A new three-step contextual simulation identifies a controller reader with a decrement interface, retaining 194 controls. Its source has 3,646 sites and its blank startup takes 57,768 transitions. This is a new source change in addition to the first composition.

For PA194C the count is

$$\underbrace{134}_{\text{fixed controller and arithmetic}} + \underbrace{67}_{\text{raw axiom readers}} - \underbrace{7}_{\text{compatible contextual folds}} = 194.$$

PA194C removes 24 primitive source sites relative to primary PA194, and has 31 fewer primitive branch outcomes. Its complete blank startup takes 57,922 transitions, versus 58,154 for primary PA194 and 58,122 for the existing A/D companion. The 232- and 200-transition differences are startup comparisons; no uniform speed improvement for all computations is asserted.

Let H_M denote the usual arithmetic sentence asserting a finite halting trace from the literal blank start of M. The proof is organized into two levels:

1. The new, machine-specific preservation result is $\mathrm{PA} \vdash H_{194C} \leftrightarrow H_{195C}$.
2. The supplied logical calibration of the unchanged numerical calculus, together with its command compiler and scheduler, yields $\mathrm{PA} \vdash (\neg H_{194C} \leftrightarrow \mathrm{Con(PA)})$.

The second statement uses inherited ordinary mathematical theorems. It does not acquire a kernel-checked status from the successful Python certificates. The current repository itself leaves its machine-specific iPRA consistency bridges open [1, 2].

2 Comparison of the entire supplied corpus

2.1 Snapshot and census scope

We inspected the requested PA tree, the incoming archives, the relevant ZFC constructions and ConciseIPRA at the pinned commit above. The 36 top-level ZIP files total 245,340,389 bytes. Recursive traversal finds 77 archive references representing 49 distinct ZIP payloads, maximum depth eight, and 26,258 file occurrences within the distinct payloads. The direct PA tree contributes 14 recognized complete tables. No archive-reading errors occurred.

The literal census recognizes numeric five-column binary tables in `.tm`, `.txt`, `.tsv` and `.csv` members. It requires exactly two rows per working control, binary writes, L/R movement and one undefined terminal target. It does not recover arbitrary tables embedded only in source code, JSON or prose. The proposed frontier machines have literal tables within this scope. Two disconnected synthetic six- and nine-state controls are excluded from the graph classification, and a no-halt synthetic control is outside the table convention. They are not proposed consistency machines.

Working states	Table occurrences	Byte forms	Action-graph classes
194	7	7	4
195	82	25	15
196	788	163	99
208	957	177	142
209	126	20	13
All recognized sizes	2,519	606	437

These are counts of artifacts, not counts of established consistency machines. They include ancestors, alternate encodings, diagnostics and explicitly rejected experiments. All definitions, member paths and hashes are in `comparison/census.json`.

For each table with a spanning root, the graph canonicalizer minimizes the read-0-first breadth-first serialization over all spanning roots. It preserves read-bit labels, write bits, movement directions and the halt target, while forgetting control names and the designated start. The counts above do not quotient by reflection or complementation. Those additional transformations are tested separately for both new 194-state tables.

All 32 archives discussed by the supplied primary PA194 report are byte-identical in the current snapshot. Their inherited contribution ledger therefore remains applicable. We newly read the four additional packets and their evidence, and freshly traversed and canonicalized all 36 archives. Appendix D gives an archive-by-archive account; the JSON and CSV ledgers retain exact hashes and the provenance of each narrative entry. This is not a new independent proof of every theorem asserted in those reports.

2.2 All existing 194-state constructions

Family	Allocation	Sites	Outcomes	Startup
Primary PA194	$134 + 67 - 7$	3,673	4,780	58,154
PA194_AD	$134 + 67 - 7$	3,673	4,780	58,122
Eight-fold PA194	$134 + 68 - 8$	3,673	4,780	58,154
New PA194C	$134 + 67 - 7$	**3,649**	**4,749**	**57,922**
New PA194CS	$135 + 67 - 8$	**3,646**	**4,745**	**57,768**

The first packet supplies primary PA194 and its A/D register-location companion [3]. The second packet supplies the eight-fold association [4]. All three retain the same separate-head 190-token axiom emitter and original command controller. Their action graphs are distinct. The two allocation patterns suggest concrete routes to 193: 67 readers with eight compatible folds, or 66 with seven. Neither route has been completed here.

The fourth existing 194-state graph is explicitly rejected. It occurs in the PA195 follow-up and has eight locally compatible folds, but its seven-zero address run is incompatible with the old return scanner. The raw runtime first fails at source ordinal 1,003, zero branch of logical register 1, after 1,309 prior outcomes. One added scanner control repairs the construction to M195S8 [8]. A correct local quotient is therefore insufficient without a correct interpreter frame.

2.3 The complete 195-state frontier

The 15 recognized 195-state graph classes split into six positive main or intermediate manuscript constructions, three auxiliary physical or search witnesses with limited endpoint claims, and six rejected diagnostics. The principal six are:

Family	Emitter/controller	Allocation	Sites	Outcomes
Original PA195	Bundled diagonal; original	$134 + 68 - 7$	3,700	4,815
195S	Separate; original	$134 + 67 - 6$	3,673	4,780
195C	Separate; cached head	$134 + 67 - 6$	3,649	4,749
Second-packet intermediate	Separate; other association	$134 + 67 - 6$	3,673	4,780
PA195DC	Bundled diagonal; cached head	$134 + 68 - 7$	3,676	4,784
M195S8	Bundled; longer scanner	$135 + 68 - 8$	3,700	4,815

The cached-head machines start their clean source computation after 57,922 literal transitions. Original PA195, 195S and the second-packet intermediate take 58,154; M195S8 takes 59,164. The original PA195 report supplies the central detailed ordinary PA calibration [5]; later packets supply the changed source, calculus and physical preservation arguments [6, 7, 8].

The second PA194 packet contains a minor report-table inconsistency: one row labels its intermediate 195-state machine as a 3,700-site diagonal layout. Its actual source, README and receipt identify a 3,673-site separate-head program with 4,780 outcomes. The table above follows those concrete artifacts.

The three auxiliary 195-state classes are a sequential-quotient search endpoint, PA195R with a parameter permutation in the restricted propositional head, and PA195N with negative diagonal vacuity. The latter two have complete physical evidence in their packet but separate, limited arithmetic-endpoint claims. The six rejected classes include incorrect two-rotation and warm/extended physical quotients, an insufficient scanner, a short-scanner deletion and an invalid zero-jump/helper fold. Their failed refinements must not be counted as successful state reductions. The detailed failure counts and source locations are preserved in `comparison/comparison.md`.

2.4 How the larger designs relate

Family	States	Structural distinction and relevance
James Chang PA209	209	Imported upstream ancestor with 3,948 source sites; distinct from the unary U209 parents.
U209/U208	209/208	Unary weak-covering theory, originally 3,909 sites. Several packets certify the same native 172/211 history fold, so their one-state savings cannot be added. The initial nine presentations form two graphs.
Expanded U208	208	Eighteen additional helper layouts bring 27 designated presentations to 20 graphs. The census's 142 classes also include exploratory and rejected tables.
Published PA196	196	3,812 sites, 4,959 outcomes and 58,154-step startup; transposed quantifier coding and revised axiom interface. The imported formal result supplies an internal implication from literal nonhalting to PA consistency.
PA196 decoder relatives	196	Association, path, scanner and register-location changes. A smaller source DAG need not yield fewer physical controls.
Closing-decrement variants	196	Four alternatives modify the same helper row. They are mutually alternative implementations, not four compatible savings.
Combined196	196	H-only Shift plus MP-to-cache: 3,788 sites and 4,928 outcomes. This controller is the source of the later 195C change.
Diagonal/Separate196	196	Diagonal vacuity in bundled or separate form gives 3,700 or 3,673 sites; these supply the logical parents of the 195/194 frontier.
Cross-archive Combined196	196	Combines those logical layouts with cached commands, giving 3,676 or 3,649 sites; zero-jump versions are additional implementations.
Encoded-negation layouts	≥ 197	An 18-site negation macro lowers one source to 3,441 sites but the retained physical implementation costs 197 controls.

The comparison therefore favors joint optimization of the numerical calculus, instruction-tree sharing, fixed controller, scanner bounds and simultaneous folds. Counting short source programs or individually valid folds alone misses essential costs.

3 A different 194-state controller: PA194C

The fallback machine M_C combines the primary 194-state packet's 67-reader, seven-fold axiom decoder with the supplied cached-head command controller. Its literal table is `construction/candidate/pa194C.tm`: 194 working states, numbered $0, \ldots, 193$, and separate halt label 194. The start is state zero on the two-way all-zero binary tape, with the head at cell zero. There are 388 transitions and no stationary moves. In native coordinates the start and halt are 74 and 254.

This is a change to the proof-search computation. Successful modus ponens stores its result in the cached head, and Shift tests that cached head alone. The old 194-state controller tests a sum involving the visible top and cached head. The new source has 3,649 primitive instruction sites, compared with 3,673 for that controller, and different command traces. A change of names for the same states would not make those changes. The axiom calculus, 190-token axiom word and its stack order are retained.

The count is

$$134 + 67 - 7 = 194.$$

The six changed native rows, relative to the primary 194-state table, are

state	read	old action	new action
38	0	$0R28$	$0R11$
38	1	$1R20$	$1R122$
164	1	$1R98$	$1R130$
214	0	$0R188$	$0R312$
214	1	$1R312$	$1R20$
312	0	$0R11$	$0R188$

All seven decoder folds are retained and checked in the resulting table.

3.1 The exact transfer theorem

Let M_{195C} denote the pinned cached-head machine in `PA195_ConPA_research_2026-11-14.zip`, and let H_M be the arithmetic sentence that M has a finite halting trace from its specified blank start. The direct preservation statement is

$$\mathrm{PA} \vdash H_{M_C} \leftrightarrow H_{M_{195C}}. \tag{C1}$$

The machine-specific content of (C1) is independently checked here; it does not rely on assuming either machine never halts.

The builder first reconstructs the pinned M_{195C} from its grammar and six folds, requiring equality of every native row with the supplied table. It then compiles the new grammar using the same cached-head fixed controller. After numbering primitive source sites from left to right, the two complete resolved instruction maps are equal: every opcode, register, successor and both decrement branches agree. Their axiom words are also equal. This latter check alone would not suffice: reassociation can change the ancestor counted by a break or continue instruction. Two independent path/interval resolvers check the stronger equality of all 3,649 resolved instructions. Hence the two source step functions are identical on every vector of nine natural registers.

A source frame has the logical register order $(N, T, W, A, D, H, X, Y, L)$, physical order map $(6, 2, 0, 4, 3, 7, 8, 5, 1)$, unary blocks $0\,1^{v+1}$, a source-address header of base length 48 plus an arbitrary nonnegative gap, and an arbitrary finite nonempty spare list. Source entry is native control 84. All 4,749 instruction outcomes, including both decrement cases, have positive-length physical simulations between the exact intended frames. The contract preserves the complete other register file and the specified spare-list effects. In particular, an increment appends a one-valued spare block; a partial-frame argument omitting this effect would be invalid.

The first checker uses the unchanged arithmetic-block contracts after checking their rows. A separate literal-tape checker rederives four list sweep lemmas from the candidate rows and checks every source outcome without an arithmetic-box oracle. Its symbolic lengths include nine independent natural register values and an arbitrary spare list. These are universal symbolic contracts, not bounded numerical simulations. The seven simultaneous folds additionally pass all 28 two-cell local contracts with arbitrary untouched exterior tape, taking at most five steps per replaced raw transition.

Both machines reach the same clean logical source seed at site 25. For M_C, an exact blank run takes 57,922 transitions and ends at native control 84, head -2, with $N = 2$, all other logical registers zero, and 26 spare blocks of value one. Independent set and bytearray simulations check the entire tape support, including markers, address bits and register blocks. There is no earlier halt.

For completeness of the transfer argument, let R_C be their common source program. Given a finite halting source trace, concatenate its physical simulation segments and prepend the verified startup to obtain a literal halting trace. Conversely, suppose a literal run first halts after T transitions. If R_C did not halt during its first $T+1$ source steps, the positive-length contracts would concatenate to a nonhalting literal run lasting more than T transitions. Determinism would contradict the first halt. Thus

$$\mathrm{PA} \vdash H_{M_C} \leftrightarrow H_{R_C}, \qquad \mathrm{PA} \vdash H_{M_{195C}} \leftrightarrow H_{R_C},$$

which proves (C1). The trace constructions use finite recursion and induction on natural loop bounds and trace lengths. Their coding and correctness are available in PA; no reflection principle is involved. The executable certificates instantiate these elementary proofs, but are not themselves a newly compiled Lean or Rocq endpoint.

3.2 Cached-head commands and their completeness

Here is the semantic reason for retaining the supplied cached controller. Put $p(a,b) = (a+b)(a+b+1)/2 + b$, $[a;w] = p(a,w)$, $I(a,b) = 4p(a,b)$, and $U(i,B) = 4p(B,i)+2$. Cantor projections are total, including $\mathrm{hd}(0) = \mathrm{tl}(0) = 0$ and $[0;0] = 0$. A command store is $\langle t,w,d,h\rangle$. With $y = d+x$, its commands are

$$S\langle t,w,d,h\rangle = \begin{cases} \mathsf{halt}, & h = 1, \\ \langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle, & h \neq 1, \end{cases}$$

$$\mathsf{MP}_x\langle t,w,d,h\rangle = \begin{cases} \langle 0,w,0,y\rangle, & I(t,y) = h, \\ \langle 0,w,y,0\rangle, & I(t,y) \neq h. \end{cases}$$

The separate-diagonal AX command pushes ten entries, including the saved top. In stack order these are

$$[q_1,\dots,q_9,t;w], \qquad (q_1,\dots,q_9) = (\Gamma(c), L_6, L_5, L_4, L_3, L_2, L_1, \rho_8(d+x,c,h+y,d), U(d,h)),$$

where $\Gamma(B) = I(B,U(B,B))$; $L_1,\dots,L_6$ and ρ_8 are the unchanged logical heads. AX returns the guarded gate as its top and clears both d and h. The heads, apart from the last universal entry, and the guarded gate all have residue zero modulo four; the universal entry has residue two. None is code one, including when the guard fails and the gate is zero.

The exact source changes bind this command interface. The successful MP tail transfers $D = y$ into initially zero H by $\mathrm{dec}\,D; \mathrm{inc}\,H; \mathrm{jump}$, then returns to fetch site 25. The transfer requires $3y+1$ primitive instructions, followed by its return. The Shift boundary has scratch $X = 0$. It clears T in $2t+1$ instructions, copies H into T and X in $4h+1$, and restores H from X in $3h+1$; thus $(T,H,X) = (h,h,0)$ after $2t+7h+3$ instructions. It sets $X = 1$ and enters the unchanged comparison/stack-decoder. These loop invariants hold for arbitrary command-store values. They justify the changed command effects independently of any finite blank-run observation.

For this ten-entry emitter define, executing left to right,

$$\begin{aligned} \mathsf{Push} &= \mathsf{AX}_{0,0,0}; S^9; \mathsf{MP}_0, \\ \mathsf{Pop} &= \mathsf{AX}_{0,0,0}; S^{11}, \\ \mathsf{Discard} &= \mathsf{MP}_1; \mathsf{AX}_{1,0,0}; S^{10}; \mathsf{MP}_0. \end{aligned}$$

Push sends any store to $\langle 0,[t;w],0,0\rangle$ without halting. Pop sends $\langle t,[a;w],d,h\rangle$ to $\langle t,w,0,a\rangle$ when $t \neq 1$ and halts when $t = 1$. Discard sends $\langle t,w,0,0\rangle$ to $\langle 0,w,0,0\rangle$ without halting. These follow from

the projection identities and residue distinctions, so they include the zero-stack alias. They must not be replaced by the nine-entry emitter's exponents.

Let Offset_0 be empty and $\mathsf{Offset}_i = \mathsf{MP}_i$ for $i > 0$. The word

$$\mathsf{Gen}_i = \mathsf{Push}; \mathsf{Offset}_i; S; \mathsf{AX}_{i,0,0}; S^{10}$$

returns $\langle U(i,a), w, 0, 0\rangle$ from $\langle a, w, 0, 0\rangle$ for every a, w, including $a = 1$. For $1 \le j \le 7$, the ordinary-head selector

$$\mathsf{Discard}; \mathsf{AX}_{A,D,H}; S^{j+1}; \mathsf{Pop}^{9-j}$$

returns q_j as top with the original stack and zero caches. The restricted head is selected, for $u \le a$, by

$$\mathsf{Discard}; \mathsf{Offset}_u; \mathsf{AX}_{b,a-u,c}; S^9; \mathsf{Pop},$$

and the gate by $\mathsf{Discard}; \mathsf{AX}_{c,b,a}; \mathsf{Pop}^{10}$. Their preserved top values are never one, making their Pop calls safe.

Modus ponens uses the suffix $S; S; \mathsf{MP}_b; S$ on

$$\langle 0, [a, I(a,b), 0; w], 0, 0\rangle.$$

It halts if $a = 1$ or $b = 1$; otherwise it returns $\langle b, w, 0, 0\rangle$. The dummy zero is essential: omitting it would consume the underlying stack. The identity remains valid for $a = b = w = 0$, despite the numerical alias of the displayed stack with zero.

Induct on a finite proof tree p in the separate-head calculus. Use the selectors for axioms and the displayed generalization word for generalization. For MP with minor proof p_0 of a and major proof p_1 of $I(a,b)$, take

$$W_p = \mathsf{Discard}; \mathsf{Push}; W_{p_1}; \mathsf{Push}; W_{p_0}; \mathsf{Push}; S; S; \mathsf{MP}_b; S.$$

The induction invariant is: from every $\langle t, w, 0, 0\rangle$, W_p either halts or returns $\langle \mathrm{concl}(p), w, 0, 0\rangle$. For a refutation p, the word

$$\mathsf{AX}_{0,0,0}; W_p; \mathsf{Pop}$$

therefore halts from *every* incoming store. The initial AX clears caches; the last Pop detects one if no earlier halt occurred. This all-store property is necessary because the unchanged exhaustive command-word scheduler carries its store between unsuccessful trials. Its enumeration of every finite command word now supplies completeness.

For soundness, annotate t, h, and every live canonical stack entry with derivations in that same numerical calculus. Zero has a fixed proof from $\rho_8(0,0,0,0) = 0$ and supplies the total stack's zero padding. AX emits axioms, a generalization of the previously derived cache, and the already derived saved top. Successful MP derives its new cache; failed MP returns zero. Shift moves derived formulas and halts only when the cache is one. Thus a halt yields a calculus refutation. This is a finite proof-store invariant, not an assertion that PA can prove the truth of all its theorems.

3.3 The inherited logical calibration and proof boundary

Write $\mathcal{C}_{\mathrm{sep}}$ for the precise separate-diagonal-head numerical calculus of the supplied 194 and 195S/195C packets. Its ordinary calibration provides primitive recursive translations of PA refutations into $\mathcal{C}_{\mathrm{sep}}$ refutations, and conversely, with PA-provable correctness. The former replaces equality-vacuity uses by diagonal-code freshness, reverse membership rewriting and neighborhood equality. The latter interprets membership by finite binary membership and obtains pointed separation from PA's formula-dependent

finite-comprehension induction. These are syntactic proof transformations; a uniform arithmetic truth predicate is not required. The reference manuscripts are retained with their provenance, rather than being presented as newly proved here.

Combining that logical calibration, the command compiler and soundness invariant above, the unchanged scheduler, and (C1) gives the ordinary mathematical conclusion

$$\mathrm{PA} \vdash (\neg H_{M_C} \leftrightarrow \mathrm{Con}(\mathrm{PA})). \tag{C2}$$

More explicitly, without importing the logical calibration, the new machine-specific result is the unconditional preservation theorem (C1); (C2) uses the supplied calibration at the stated interface. This work does not convert the existing conditional Lean/iPRA arithmetic endpoints into complete formal proofs. No newly kernel-checked end-to-end PA, iPRA, Lean, or Rocq theorem is claimed. The new finite and universal symbolic checks establish the decoder/controller composition and identify its exact literal table; the logical PA calibration remains an inherited ordinary mathematical proof.

4 A second 194-state machine from a symmetric guard

The second new machine, M_{CS} or PA194CS, changes the source computation of Shift and then folds the new reader into an existing decrement control. It has 194 working states and 388 transition rows, with the same blank binary one-tape convention as PA194C. Its source has **3,646 sites** and **4,745 primitive outcomes**. The literal table is `secondary/pa194CS/machine/pa194CS.tm`; the dense SHA-256 is

`1c71002e5b8a2ddd1dd1311a2c54c424764624b611b69c8efbee1ac23c2e6f5f`.

This is an additional source simplification. It retains the exact cached command interface of PA194C, while removing an unnecessary restoration loop from its implementation.

4.1 The comparison routine is symmetric in its full effect

The first source instructions of both programs are the following finite register graph. Here $H = r_5$, $X = r_6$, and a decrement's first successor is its zero branch.

0	dec H $[4, 1]$
1	dec X $[3, 2]$
2	jump 0
3	jump 5
4	dec X $[10, 5]$
5	dec H $[7, 6]$
6	jump 5
7	dec X $[9, 8]$
8	jump 5
9	jump 11
10	halt.

From any natural pair $(H, X) = (a, b)$ this routine halts exactly when $a = b$. If $a \neq b$, it reaches site 11 with $H = X = 0$, preserving every other register. To prove this, the loop $0, 1, 2$ removes one from each positive component. If H first becomes zero, instruction 4 distinguishes equality from a positive remainder in X. If X first becomes zero while H is positive, instruction 3 enters cleanup. Sites $5, 6$

clear the remaining H, and sites $7, 8$ clear the remaining X; the latter loop may revisit the already-zero H and still terminates. Induction on $a + b$ establishes these statements in PA.

Thus exchanging (a, b) with (b, a) preserves both halting and the entire surviving register store. Equality of the Boolean test alone would not suffice for this optimization; clearing *both* comparison registers is the property needed by the following stack decoder.

4.2 The shortened Shift

At the inherited Shift boundary let $(T, H, X) = (t, h, 0)$. PA194C performs

$$\begin{aligned}(t, h, 0) &\longmapsto (0, h, 0)\\ &\longmapsto (h, 0, h)\\ &\longmapsto (h, h, 0)\\ &\longmapsto (h, h, 1),\end{aligned}$$

then enters the comparison. PA194CS instead performs only

$$(t, h, 0) \longmapsto (0, h, 0) \longmapsto (h, 0, h) \longmapsto (h, 1, h)$$

and enters that same comparison. The deleted loop restores X into H; the final increment has changed from X to H. By the preceding lemma, both computations halt exactly when $h = 1$. Otherwise both reach the unchanged stack decoder at site 11 with

$$(T, H, X) = (h, 0, 0).$$

The unchanged decoder therefore produces the same command store in both machines. This holds for every natural t, h, including $h = 1$; there is no consistency assumption in the argument.

In fact, the source-block equivalence is stronger than the command-boundary claim. For arbitrary incoming scratch value x, the common clear/copy prefix sends $(T, H, X) = (t, h, x)$ to $(h, 0, k)$ with $k = h + x$. The old tail and guard then compare $(H, X) = (k, 1)$, while the new ones compare $(1, k)$. They again have identical halting and complete surviving-register effects. Thus this local source proof requires no zero-scratch hypothesis.

The finite source binding is exact. The first 3,644 resolved instructions of the new source agree with those of the pinned cached controller. The old final five instructions are

3644	dec X [3647, 3645]
3645	inc $H \to 3646$
3646	jump 3644
3647	inc $X \to 3648$
3648	jump 0,

whereas the new tail is

$$3644 : \text{inc}\, H \to 3645, \qquad 3645 : \text{jump}\, 0.$$

The preceding transfer loop's zero exit remains site 3644 and now enters the shorter tail. An incoming-edge check shows that no instruction outside the replaced tail enters its interior: the only external edge to the tail is the zero branch from site 3640, which establishes $H = 0$. Consequently the common source can be partitioned into unchanged single steps and complete shortened-Shift/comparator blocks. The blocks terminate on every natural register store, have the same halt outcome or the same full store at site 11, and have positive duration. Induction on these blocks gives halting equivalence of the two

source programs from site 25 for arbitrary initial registers. This is a direct local source simulation; the entire source graphs are not asserted equal.

All AX formulas, command selectors, MP code and scheduler instructions are retained. The logical registers agree at command boundaries; physical spare lists need not have equal lengths because the number of increments changes. Each machine's complete-frame simulation tracks its own spare-list effects, so this difference causes no omission in the tape argument.

4.3 Recovering the state cost by an exact contextual fold

A direct physical implementation uses one new reader, native control 330:

$$(330, 0) : 0R63, \qquad (330, 1) : 1L73.$$

Relative to PA194C, redirect $(214, 1)$ to $1R30$ and $(310, 1)$ to $1R330$. This produces a 195-state intermediate for the shortened source. Control 41, with actions $0R24$ and $0R25$, is entered only through $(28, 0) : 0R41$ or $(70, 0) : 0R41$. The new 330 is entered only through $(310, 1) : 1R330$. Consequently the preceding tape bit distinguishes the two controls: zero for 41 and one for 330.

Identify 41 with 330, replace the retained rows by

$$(330, 0) : 0L31, \qquad (330, 1) : 1L46,$$

and redirect all old targets 41 to 330. The following exact three-step paths reproduce the old one-step actions. The intermediate head visits only the current and preceding tape cells.

preceding	current	old action	controls along replacement
0	0	$0R24$	$330 \to 31 \to 21 \to 24$
0	1	$0R25$	$330 \to 46 \to 61 \to 25$
1	0	$0R63$	$330 \to 31 \to 47 \to 63$
1	1	$1L73$	$330 \to 46 \to 62 \to 73$

For every row, the preceding bit and all exterior tape are preserved, and the final current bit, head coordinate and target agree with the old action. These four cases and every incoming row are checked against the *final* quotient table. Ordinary transitions commute with the projection. Induction gives an all-tape positive simulation from the specified start; the usual determinism argument gives the converse halting implication. The original seven AX folds remain separately verified. Counting from the fully unfolded control table gives

$$135 + 67 - 7 - 1 = 194.$$

4.4 Consistency calibration and verification scope

The changed Shift has exactly the cached command semantics already proved for PA194C. Its MP and AX commands are unchanged. Consequently the same ten-entry proof compiler, all-store universal refutation word, proof-store soundness invariant and exhaustive scheduler apply without changing a stack exponent or an axiom schema. Induction on command executions gives the same source refutation criterion. The guard proof above uses only natural-number loop induction; the finite source binding and contextual quotient use primitive recursive trace transformations. All are formalizable in PA.

The completed literal table is checked on all 4,745 universal source outcomes, with arbitrary natural primary-register values, both decrement branches, an arbitrary gap and an arbitrary finite nonempty spare list. The independent checker rederives the same four list-sweep lemmas from literal transitions,

resolves the new source graph twice, and verifies the exact shorter tail against the pinned cached source. It does not assert source-graph identity with PA195C, which would be false.

The *folded* machine reaches the exact complete clean source frame after **57,768** blank-tape transitions. The 195-state intermediate takes 57,594; those counts refer to different tables. Both reach site 25, native control 84, head -2, logical seed $(2, 0, 0, 0, 0, 0, 0, 0, 0)$ and the recorded complete spare list. Thus startup, source semantics and full physical simulation all refer to the delivered 194-state table.

With the same explicitly inherited ordinary syntactic calibration of the separate-head numerical calculus, these facts give

$$\mathrm{PA} \vdash \big(\neg H_{M_{CS}} \leftrightarrow \mathrm{Con}(\mathrm{PA})\big).$$

The new contribution is the symmetric-guard source proof and its exact 194-state implementation. The logical PA calibration is inherited; no new end-to-end proof-assistant kernel theorem is asserted.

Appendix C gives an independent expanded induction proof of the symmetric guard, its exact source-step costs and the global source binding. In particular, writing $k = h + x$, the complete shortened block saves zero source steps at $k = 0$, four at $k = 1$, and $2k + 4$ at $k \geq 2$. These are source-step identities, not a uniform literal-tape runtime comparison.

5 Why the symbolic physical checks are universal

The independent checker represents complete tapes by finite sequences of fixed bits, constant-bit runs with affine lengths, arbitrary register blocks and an arbitrary finite spare list, with blank exterior on both sides. Its affine coordinates include the gap, nine independent register values, the first spare value and the remaining list width. A positive decrement represents the tested value as $v + 1$; its zero branch sets that value to zero. Other values remain independent.

Each elementary certificate rule has an ordinary arithmetic proof:

1. A literal transition is applied only when the scanned run has a positive constant part, so the scanned bit is determined for every assignment of its natural parameters.

2. A same-control sweep through a constant-bit run is exactly repeated application of one transition, proved by induction on its length. Its time and displacement are recorded symbolically.

3. A list sweep is justified by a generic one-block trace with an arbitrary value and arbitrary exterior contexts, followed by induction on the finite list. The four used one-block traces are rederived from the delivered table; each has duration $v + 2$.

4. Normalization uses exact concatenation identities. For example, if $B(v) = 0\,1^{v+1}$ and $C(v) = 1\,0\,1^{v}$, then $C(v)1 = 1B(v)$. Its extension over a list and its reversal follow by finite-list induction. Adjacent blank zero runs may be absorbed into the infinite blank exterior.

Induction on the checked certificate establishes equality with the intended complete target frame. Nonnegative affine coefficients certify all-natural length bounds. The positive constant part of the duration is checked separately; the minimum over the delivered source contracts is three. A nonhalting source outcome closes only at its exact target frame, and cannot pass through the terminal control on the way.

These rules explain the universal quantification behind the 4,749 and 4,745 checks. The recorded 1,096,302 and 1,100,204 symbolic steps are verifier work, not a bound on the physical runtime for arbitrary register values. The finite 57,922- and 57,768-step blank runs serve a different purpose: they

establish the exact initial frames, including every nonzero tape cell. They are not evidence obtained by extrapolating a long nonhalting prefix.

5.1 Verification results

Check	PA194C	PA194CS
Working states / transition rows	194 / 388	194 / 388
Resolved source instructions	3,649	3,646
Source graph identical to 195C	Yes	No: proved block change
AX fold cases / additional controller cases	28 / 0	28 / 4
Maximum local fold duration	5	5
Construction checker source outcomes	4,749	4,745
Construction checker symbolic steps	1,507,397	1,513,513
Independent checker source outcomes	4,749	4,745
Independent checker symbolic steps	1,096,302	1,100,204
Independent maximum per outcome	403	403
Minimum positive duration constant	3	3
Generic list lemmas rederived	4	4
Exact complete-tape blank startup	57,922	57,768
Novel among existing 194 graphs	Yes	Yes
New end-to-end proof-assistant endpoint	No	No

The construction checker additionally checks the unchanged arithmetic-box rows. The independent checker imports neither the compiler nor an arithmetic-box oracle. Two independent control-flow resolvers agree on every successor. The recorded JSON evidence includes the exact hashes, startup support, source maps, all outcome durations and list-lemma traces.

6 Logical calibration and its exact dependency

The unchanged numerical calculus codes implication, membership, universal quantification and equality by

$$I(A,B) = 4p(A,B), \quad M(a,b) = 4p(a,b)+1, \quad U(a,B) = 4p(B,a)+2, \quad E(a,b) = 4p(a,b)+3,$$

where $p(a,b) = (a+b)(a+b+1)/2 + b$. Code zero is truth, code one is $M(0,0)$, and $N(B) = I(B,1)$. The transposed quantifier arguments are part of the exact definition. The separate diagonal head is $\Gamma(B) = I(B, U(B,B))$.

The supplied calibration uses the strict descent of formula components to show that variable label B is absent from the formula coded by B. The exceptional truth case and $I(0,0) = 0$ are treated separately. Thus diagonal vacuity has a primitive-recursive proof-producing translation into ordinary first-order arithmetic. The bundled and separate diagonal presentations are intertranslatable by fixed classical implicational derivations and suitable fresh parameters.

For the direction from numerical refutations to PA refutations, membership is interpreted as finite binary membership. One convenient formula is

$$\mathrm{Bit}(s,z) \equiv \exists u, v\, [s = (2u+1)2^z + v \ \wedge \ v < 2^z].$$

For each arithmetic formula θ, PA proves finite binary comprehension by induction on the bound: retain the old bitset when the new instance of θ fails, and add 2^n when it holds. Substitution of a formula's syntax into that induction-proof template is a primitive recursive operation on proof codes. This is a syntactic construction, not an appeal to an arithmetic truth predicate. Pointed separation uses this comprehension with the proper variable-collision cases. PA proves $\neg\,\mathrm{Bit}(x, x)$ uniformly, since $\mathrm{Bit}(s, z)$ implies $z < s$. Thus a translated proof of code one, the atom $M(0, 0)$ with variable label zero, yields a PA refutation.

For the converse, the supplied proof recovers reverse membership rewriting and neighborhood equality using pointed separation and diagonal-code freshness, then substitutes these interfaces into the PA-refutation compiler. The relevant proofs and exact axiom heads are retained in `construction/logical_reference/`, including the original PA195 calibration, the supplied 195S/195C preservation and the primary PA194 calibration. They are cited inherited results rather than a fresh claim that the physical verifier proves a logical theorem.

The command proof in Section 3 establishes both directions between the unchanged calculus and the new source controller. It handles the ten-entry AX interface, the zero-stack alias, early halts, and a refutation word that works from every incoming store. The latter is essential because the scheduler carries its store between trials. The physical proof then transports that source result to the exact literal binary machine. All proof-store annotations, finite trace concatenations and bounded source iterations are available in PA.

Layer	Evidence used	Status here
Literal tables and sources	Complete rows, two CFG resolvers, parent binding and symmetric-block proof	Fresh finite verification and ordinary proof
Physical simulation	Arbitrary-register contracts, list inductions, exact startup	Fresh executable certificates and ordinary arithmetic proof
Cached command calculus	Explicit selectors, proof compiler, all-store invariant	Ordinary proof, same supplied command convention
PA logical calibration	Uniform syntactic refutation translations for the separate-head calculus	Inherited ordinary mathematical dependency
End-to-end Lean/Rocq theorem	Would assemble all layers for this exact table	Not supplied by this work

There is no use of Con(PA) as an extra axiom in the claimed PA biconditional. Actual nonhalting follows if PA is consistent. The conditional iPRA endpoints in the repository remain conditional, and external nonhalting in a proof assistant's metatheory must not be confused with an internal PA consistency implication.

7 Search for fewer than 194 controls

No verified machine below 194 controls was found. The searches were frozen after the following recorded work. They are discovery attempts within specified transformation families, not lower-bound proofs.

7.1 Decoder allocation and contextual quotients

The main fitness jointly counts shared axiom readers and compatible history-sensitive folds, subject to the source scanner constraints. We seeded it from both existing 194-state associations, with either the original or cached controller. Variations included shared-subtree rotations, single-occurrence rotations,

multiple rotations, bijections of the three parameter roles within universally instantiated axiom heads, and axiom-head ordering. No new logical transformation is needed for exact-word rotations. A successful parameter/order variant would also need a correctly rebound source and axiom-selection proof; a favorable fitness value would not suffice.

Seven stochastic runs executed at least **24,295,755 mutation-proposal slots**. This is not a count of distinct trees or evaluated machines: proposals can repeat or be rejected before full scoring. Two runs completed their stopping conditions; five were stopped at the recorded checkpoint, and their lower counts are taken from the last complete logged cycle. All retained best scores were 60 effective readers, hence 194 total controls with the 134-control fixed part.

A deterministic neighborhood traversal expanded 5,000 distinct roots and evaluated 1,052,340 neighboring rotations. It admitted grammars with scanner penalty zero and effective-reader score at most 60. There were 587 roots still queued at the cap, so this traversal was *not* exhaustive. A separate fixed-shape screen examined all 720 orders of six heads on each of the two baseline shapes, 1,440 templates in total, without improvement.

We also closed three raw tables under the specified general local pair-contraction rule. It uses abstract reachable 11-cell windows (radius five), allows replacement first actions, and requires local rejoining within 150 transitions. The finite closure queues exhausted:

Parent	Initial controls	Distinct labeled tables	Best
Primary raw table	201	192	194
Second raw table	202	640	194
Cached primary raw table	201	192	194

The completeness here is only the closure under that exact bounded rule and those seeds. It says nothing about larger local windows, longer simulation segments, joint changes outside that rule or other sources.

To avoid testing only the two archived endpoints, a further pipeline generated 1,000 grammar files, representing 606 distinct raw tables. Each was reduced to 194 controls with an explicit, simultaneously checked fold witness, then passed to the broader 11-cell/150-step action merger. None yielded another contraction. These sampled tables were search witnesses; no blanket claim of their complete blank-start consistency calibration is made.

7.2 Whole-controller screening

We separately collected symbolic local tape contexts from every source outcome and the exact startup of primary PA194 and PA194C. The first screens retained the arithmetic-box premises and considered the 171 other controls, with radius five and local budgets of 150–200 steps. Neither supplied a contraction.

The final PA194C screen removed that arithmetic-state exclusion. It replayed the independent literal-tape proof, collected contexts at all 194 controls, included the interiors skipped by unary sweeps, included the generic arbitrary-context one-block proofs used for list sweeps, and included the full blank startup. It recorded 18,976 distinct local views. Both a direct redirection filter and a general common-first-action pair-merger filter, with radius five and 300-step local budgets, returned no candidate. The context collector deliberately overapproximates unknown lengths for screening. Any surviving edit would still have required a new complete physical certificate.

The portable programs, seeds and frozen discovery logs are under `search/decoder/`; the controller collectors, local filters and receipts are under `search/controller/`. Their JSON summaries state exact scopes and interruption boundaries.

7.3 Further source-level work

One sound acceptance extension is to halt when the cached formula code has residue one modulo four. Such a code is a membership atom $M(a, b)$. A PA proof of its arithmetic translation yields a refutation by substituting zero for its free variables, since PA refutes $\mathrm{Bit}(0, 0)$. Conversely, contradiction code one is included, so the old completeness route remains available. A short source guard can perform the residue test after the cache-copy loop. Its straightforward physical implementation introduced new reader controls while the old equality guard was heavily shared; no state reduction was obtained.

A second idea preserves the cached Shift interface exactly: after clearing T and transferring H into both T and X, the values are $(T, H, X) = (h, 0, h)$. Set $H = 1$ and use the symmetric equality test against X, avoiding restoration of H and the old $X = 1$ setup. This reduces the source from 3,649 to 3,646 sites. An attempted direct reassociation failed a universal runtime contract, but adding a reader and then applying the new $41 \mapsto 330$ contextual fold succeeded. This is the accepted PA194CS construction proved in Section 4.

A final targeted search explored the symmetric implementation. From its 195-state intermediate, the declared radius-five/300-step local rule reached three tables, including two 194-state endpoints, and exhausted the queue. Allowing arithmetic states as merge endpoints gave the same count. Starting instead from its fully unfolded 202-state table and allowing every working control except the start, the search explored 576 tables, including four 194-state endpoints, and exhausted the queue. The original seven AX folds were rebound to that raw parent. No 193-state table was found. These separate records are under `search/symmetric_shift/`; the earlier frozen search counts are unchanged.

The successful fallbacks are PA194C and PA194CS. The failed searches do not imply that their count is optimal, and none is used as a premise of their proofs.

8 Ideas transferred from ZFC and ConciseIPRA

The most concrete ZFC connection is the cached-result return convention. The K217 cache-return composition explicitly moves a successful modus-ponens conclusion into the cache instead of pushing a dummy zero and copying to the current-formula register [10]. Its source and decoder checks did not themselves save a physical control, and its modified full runtime had a separately stated proof boundary. The PA corpus later combines the corresponding convention with H-only Shift. PA194C carries that established PA controller into the smallest existing decoder allocation and supplies the complete changed physical checks.

The wider ZFC search also illustrates why source simplification and machine minimization must be combined: register roles, marker history, scanner range and available helper states determine whether two apparently useful savings can coexist [11]. We used this principle in the reader/fold searches and in verifying the controller composition in its final table.

ConciseIPRA's small kernels derive substantial logical infrastructure from a small native interface, and its artifacts insist on complete checker costs and explicit interpretation boundaries [12]. That suggests looking for derived axiom heads and alternative proof objects. It does not license replacing the PA proof search by an iPRA proof search: their consistency assertions are different objectives. Nor does a smaller Python checker automatically give a smaller binary one-tape machine. This construction retains the existing PA-calibrated calculus and uses those methodological lessons without attributing a new logical compression to ConciseIPRA.

9 Reproduction and package contents

All primary construction and verification commands use Python 3.10 or later and its standard library. Assertions must be enabled. From the archive root run:

```
python3 -B construction/build_cached.py
python3 -B construction/verification/check_cached.py
python3 -B comparison/verify_novelty.py \
  construction/candidate/pa194C.tm \
  --census comparison/census.json
```

The first command regenerates the exact table and the construction receipt. The second writes new independent results under `construction/verification/results/`. It can take several minutes. The third checks the finite corpus novelty claim. Neither the stochastic discovery search nor a new repository download is necessary to rebuild and verify the machine.

For the symmetric-guard companion, run:

```
python3 -B secondary/pa194CS/build.py
python3 -B secondary/pa194CS/verification/check.py
python3 -B comparison/verify_novelty.py \
  secondary/pa194CS/machine/pa194CS.tm \
  --census comparison/census.json
```

The first two commands reconstruct the shorter source and exact fold, and independently verify every physical outcome and the complete seed. The additional novelty receipt also compares PA194CS with PA194C.

To rerun the entire corpus census, supply a local checkout at the pinned commit with the PA and incoming paths available:

```
python3 comparison/census.py --repo /path/to/ProveIt \
  --output /path/to/new-census
```

The script refuses another commit. The recorded census already includes all paths and hashes. A separate selected-machine CSV preserves the earlier report's positive and experimental evidence classifications.

The full LaTeX source is in `report/`. Rebuild with two runs of `pdflatex -interaction=nonstopmode -halt-on-errorpa194_cached_research.tex` from that directory. The archive's `SHA256SUMS` binds the delivered files. Regenerated timing receipts may differ from recorded receipts; the transition table is deterministic and must have the published hash.

A Complete PA194C transition table

Each action is the written bit, L/R motion and next control. The symbol H is the separate terminal control 194. The initial control is zero. This appendix is generated directly from the delivered dense table.

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
0	1R1	1R2	65	0R88	1R30	130	0R134	1R135
1	1R3	1R1	66	0R89	1R66	131	0R136	1R137
2	0R0	1R2	67	0L90	1L33	132	0R138	1R139
3	0R4	0R1	68	0L81	0R2	133	0R140	1R141
4	1R5	1L6	69	0R91	1R69	134	0R97	1R35
5	1L7	1L8	70	0R72	1R92	135	0R142	1R124
6	0L7	1L6	71	0R46	1R93	136	0R143	1R144
7	0L9	1L6	72	0R42	0L13	137	0R145	1R146
8	0L10	0L11	73	0R94	1R37	138	0R134	1R147
9	1R12	0L9	74	0R30	1R95	139	0R88	1R148
10	0L9	0L10	75	0R42	1R96	140	0R149	1L131
11	1L8	1L11	76	0R97	1R98	141	0R150	1R151
12	0R12	0R13	77	0R99	1L100	142	0R76	1R41
13	0L14	0R13	78	0R30	1R77	143	0R28	1R152
14	1L15	0L15	79	0R93	1R101	144	0L153	1L154
15	0L15	1L16	80	0R30	1R102	145	0L129	1R155
16	0L17	1L16	81	1R81	1L15	146	0R156	1R157
17	0L18	1L16	82	0R103	1R82	147	0R149	1R158
18	0L19	1L16	83	0R94	0L7	148	0R159	1R88
19	0L20	1L16	84	0R104	1R105	149	0R97	1R160
20	0L21	1L16	85	0R42	1R106	150	0R161	1R162
21	0L22	1L16	86	0R107	1R48	151	0R163	0L10
22	0L23	1L16	87	0R108	0L72	152	0R164	1R165
23	0R23	1R24	88	0R68	0R66	153	0R166	1R157
24	0R25	1R26	89	0R109	1R89	154	0R167	1R156
25	0R27	1R28	90	0R49	1R100	155	0R135	1R130
26	0R29	1R30	91	0R110	1R91	156	0R147	1R41
27	0R31	1RH	92	0R30	1R48	157	0R134	1R168
28	0L32	1R33	93	0R47	1R60	158	0R41	1L169
29	0R34	1R35	94	0L63	0R89	159	0L170	1R171
30	0L36	1L37	95	0R94	1L111	160	0R107	1R85
31	0R38	1R39	96	0R101	0L14	161	0R172	1R170
32	0R40	1R41	97	0R36	1R112	162	0R173	1R174
33	0R42	1R43	98	0R111	1R75	163	0R175	1R176
34	0R44	1R45	99	0L63	0R109	164	0R177	1R178
35	0R46	1R33	100	0L62	1R113	165	0L179	1L180
36	0R47	1R48	101	0L90	1R114	166	0R149	1R140
37	0R49	1R50	102	0R115	0L14	167	0R165	1R181
38	0R51	1R42	103	0R69	1R103	168	0L130	1R88
39	0R52	1R53	104	0R116	1R95	169	0R182	1R135
40	0R54	1R55	105	0R116	1R67	170	0R183	1R145
41	0R56	1R57	106	0R117	0L14	171	0R158	1R130
42	0R58	0R59	107	0R118	1L71	172	0R139	1R165
43	0R60	0L14	108	0R119	1R120	173	0R158	1R184
44	0R61	1R62	109	0R14	1R109	174	0R185	1R107
45	0R63	1R52	110	0R66	1R110	175	0R31	1R52
46	0R64	1R65	111	0R121	1L35	176	0R186	1R175
47	0R58	0R66	112	0R122	1R123	177	0R134	1R142
48	0R67	0L14	113	0R88	0L14	178	0R32	1R187
49	0R68	0R69	114	0R90	1L95	179	0R144	1R188
50	0R70	0L48	115	0R124	1R124	180	0R189	1R190
51	0R64	1R53	116	0L63	0R103	181	0R41	1R191
52	0L71	0L15	117	0R68	0R91	182	0R168	1R41
53	0R72	0L10	118	0R58	0R91	183	0R192	1R88
54	0R73	1R74	119	0L86	1R125	184	0R153	1R146
55	0R75	1R76	120	0R126	1R127	185	0R165	1R180
56	0R77	1R78	121	0R58	0R110	186	0R92	1R111
57	0R79	1R80	122	0R99	1R113	187	0R193	1R159
58	0L81	0R1	123	0R79	1R78	188	0R169	1R154
59	0R82	1R59	124	0R88	1R88	189	0R158	1R192
60	0L73	1L46	125	0R128	1R129	190	0R179	1R182
61	0R83	1R84	126	0R32	1R36	191	0R182	1R41
62	0R54	1R85	127	0R31	1R111	192	0R149	1R168
63	0L86	1R87	128	0R130	1R131	193	0R76	1R144
64	0R58	0R82	129	0R132	1R133			

B Complete PA194CS transition table

The action convention, initial state zero and separate halt label 194 are the same as in Appendix A. This table is generated directly from `secondary/pa194CS/machine/pa194CS.tm`.

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
0	1R1	1R2	65	0R85	1R86	130	0R134	1R135
1	1R3	1R1	66	0R35	1R87	131	0R136	1R137
2	0R0	1R2	67	0R41	0L13	132	0R138	1R139
3	0R4	0R1	68	0R88	1R65	133	0R140	1R141
4	1R5	1L6	69	0R47	1R89	134	0R91	1R120
5	1L7	1L8	70	0R41	1R90	135	0R142	1R123
6	0L7	1L6	71	0R91	1R92	136	0R143	1R144
7	0L9	1L6	72	0R93	1L94	137	0R145	1R146
8	0L10	0L11	73	0R47	1R72	138	0R134	1R147
9	1R12	0L9	74	0R87	1R95	139	0R46	1R148
10	0L9	0L10	75	0R47	1R96	140	0R149	1L131
11	1L8	1L11	76	1R76	1L15	141	0R150	1R151
12	0R12	0R13	77	0R88	0L7	142	0R71	1R40
13	0L14	0R13	78	0R97	1R98	143	0R28	1R152
14	1L15	0L15	79	0R41	1R99	144	0L153	1L154
15	0L15	1L16	80	0R100	1R84	145	0L129	1R155
16	0L17	1L16	81	0R101	0L67	146	0R156	1R157
17	0L18	1L16	82	0R102	1R82	147	0R149	1R158
18	0L19	1L16	83	0R103	1R83	148	0R159	1R46
19	0L20	1L16	84	0R104	0L14	149	0R91	1R160
20	0L21	1L16	85	0R62	0R102	150	0R161	1R162
21	0L22	1L16	86	0R105	0L84	151	0R163	0L10
22	0L23	1L16	87	0R30	1R57	152	0R164	1R165
23	0R23	1R24	88	0L60	0R83	153	0R166	1R157
24	0R25	1R26	89	0R88	1L106	154	0R167	1R156
25	0R27	1R28	90	0R95	0L14	155	0R135	1R130
26	0R29	1R30	91	0R64	1R107	156	0R147	1R40
27	0R31	1RH	92	0R106	1R70	157	0R134	1R168
28	0L32	1R33	93	0L60	0R103	158	0R40	1L169
29	0R34	1R35	94	0L59	1R108	159	0L170	1R171
30	0L33	1L36	95	0L109	1R110	160	0R100	1R79
31	0R37	1R38	96	0R111	0L14	161	0R172	1R170
32	0R39	1R40	97	0R112	1R89	162	0R173	1R174
33	0R41	1R42	98	0R112	1R104	163	0R175	1R176
34	0R43	1R44	99	0R113	0L14	164	0R177	1R178
35	0R45	1R36	100	0R114	1L66	165	0L179	1L180
36	0R46	1R47	101	0R115	1R116	166	0R149	1R140
37	0R48	1R41	102	0R117	1R102	167	0R165	1R181
38	0R49	1R50	103	0R14	1R103	168	0L130	1R46
39	0R51	1R52	104	0L109	1L33	169	0R182	1R135
40	0R53	1R54	105	0R67	1R118	170	0R183	1R145
41	0R55	0R56	106	0R119	1L120	171	0R158	1R130
42	0R57	0L14	107	0R121	1R122	172	0R139	1R165
43	0R58	1R59	108	0R46	0L14	173	0R158	1R184
44	0R60	1R49	109	0R85	1R94	174	0R185	1R100
45	0R55	0R61	110	0R109	1L89	175	0R31	1R49
46	0R62	0R63	111	0R123	1R123	176	0R186	1R175
47	0L64	1L65	112	0L60	0R82	177	0R134	1R142
48	0R45	1R50	113	0R62	0R117	178	0R32	1R187
49	0L66	0L15	114	0R55	0R117	179	0R144	1R188
50	0R67	0L10	115	0L80	1R124	180	0R189	1R190
51	0R68	1R69	116	0R125	1R126	181	0R40	1R191
52	0R70	1R71	117	0R127	1R117	182	0R168	1R40
53	0R72	1R73	118	0R47	1R84	183	0R192	1R46
54	0R74	1R75	119	0R55	0R127	184	0R153	1R146
55	0L76	0R1	120	0R35	1R33	185	0R165	1R180
56	0R61	1R56	121	0R93	1R108	186	0R118	1R106
57	0L68	1L35	122	0R74	1R73	187	0R193	1R159
58	0R77	1R78	123	0R46	1R46	188	0R169	1R154
59	0R51	1R79	124	0R128	1R129	189	0R158	1R192
60	0L80	1R81	125	0R32	1R64	190	0R179	1R182
61	0R82	1R61	126	0R31	1R106	191	0R182	1R40
62	0L76	0R2	127	0R63	1R127	192	0R149	1R168
63	0R83	1R63	128	0R130	1R131	193	0R71	1R144
64	0R30	1R84	129	0R132	1R133			

C Independent proof of the symmetric destructive guard

The symmetric-guard 194-state variant is materially different from the cached-controller 194-state machine: its primitive program has 3,646 rather than 3,649 instructions. The following proof checks its precise source-level preservation. The physical refinement and additional state fold are separate obligations, checked in the construction package.

A decrement instruction $\mathrm{dec}(r, z, p)$ jumps to z when register r is zero; otherwise it decrements r and jumps to p. Logical registers H, X have indices 5 and 6. The common source prefix is exactly

0	$\mathrm{dec}(H, 4, 1)$	1	$\mathrm{dec}(X, 3, 2)$
2	$\mathrm{jump}(0)$	3	$\mathrm{jump}(5)$
4	$\mathrm{dec}(X, 10, 5)$	5	$\mathrm{dec}(H, 7, 6)$
6	$\mathrm{jump}(5)$	7	$\mathrm{dec}(X, 9, 8)$
8	$\mathrm{jump}(5)$	9	$\mathrm{jump}(11)$
10	halt.		

Let $G(a, b)$ count source instructions until the first entry to control 10 or 11 from control zero and $(H, X) = (a, b)$. It does not count the halt instruction itself. Every other register is arbitrary.

Guard lemma. The computation terminates at control 10 exactly when $a = b$, and otherwise at control 11. In both cases $H = X = 0$, and every other register is unchanged. Its exact duration is

$$G(a, b) = \begin{cases} 3a + 2, & a = b, \\ 2a + b + 4, & a > b, \\ 3b + 2, & a < b. \end{cases} \tag{SG1}$$

Proof. If both registers are positive, instructions $0, 1, 2$ consume exactly one unit from each and return to control zero in three steps. Induction therefore reduces the pair to one with a zero component after $3\min(a, b)$ steps. If both are zero, the two zero branches at 0 and 4 reach control 10.

The cleanup loop at control 5 has a separate exact invariant. With $(H, X) = (u, v)$, each positive H costs two steps $(5, 6)$ and reduces H by one. Once $H = 0$, each positive X costs three steps $(5, 7, 8)$ and reduces X by one. The final path $5, 7, 9$ exits at 11. Thus cleanup always clears both registers in $2u + 3v + 3$ steps. This statement follows by induction first on u and then on v.

If $a > b$, after the paired decrements the path $0, 1, 3$ takes three steps to cleanup with $(H, X) = (a - b - 1, 0)$. The total is

$$3b + 3 + 2(a - b - 1) + 3 = 2a + b + 4.$$

If $a < b$, the path $0, 4$ takes two steps to cleanup with $(H, X) = (0, b - a - 1)$. The total is

$$3a + 2 + 3(b - a - 1) + 3 = 3b + 2.$$

All inspected instructions mention only H and X, proving the full register-frame statement. This establishes (SG1) for all natural inputs. In particular the outcome and complete exit register vector are invariant under exchanging the two guard inputs, although the duration can change.

Application to the changed Shift tail. Enter source control 3,638 with $(T, H, X) = (t, h, x)$ and arbitrary other registers. The common prefix clears T, then transfers each unit of H into both T and X. It uses

$$(2t + 1) + (4h + 1) = 2t + 4h + 2$$

steps and leaves $(T, H, X) = (h, 0, k)$, where $k = h + x$. The old tail restores X to H, sets $X = 1$, and jumps to control zero; it uses $3k+3$ further steps and presents guard inputs $(k, 1)$. The new tail merely increments H and jumps to zero; it uses two steps and presents $(1, k)$. By the guard lemma, the two source computations have exactly the same exit control and full register vector. They both halt when $k = 1$; otherwise they continue at source 11 with $H = X = 0$. No assumption $x = 0$ is needed for this preservation lemma. The normal cached-command interpretation uses the inherited $x = 0$ command boundary, where $k = 1$ is precisely the intended test $h = 1$.

The total old and new durations to first entry at 10 or 11 are

$$D_{\text{old}} = 2t + 4h + 3k + 5 + G(k, 1), \qquad D_{\text{new}} = 2t + 4h + 4 + G(1, k).$$

Consequently

$$D_{\text{old}} - D_{\text{new}} = \begin{cases} 0, & k = 0, \\ 4, & k = 1, \\ 2k + 4, & k \geq 2. \end{cases} \tag{SG2}$$

Counting the final halt instruction would add one to both equal-case durations and leave (SG2) unchanged. This is a source-step comparison; it is not a claim of a uniform literal-tape speed improvement.

Binding and global source preservation. The complete old and new maps agree at all controls $0, \ldots, 3643$. The old five-instruction tail at 3644–3648 is replaced by the two instructions $\text{inc}(H, 3645)$ and $\text{jump}(0)$. The only entry from outside the Shift block 3638–3648 is the zero branch from source 48 to source 3638; no outside instruction enters the removed tail. These are exact finite properties checked by `check_symmetric_source.py`, including every successor. Thus equal configurations outside the changed block synchronize either by the identical next instruction or by the finite positive block simulation just proved. A block ending at 10 is followed by the same halt instruction. Induction on synchronized segments and the bounded trace argument therefore give halting equivalence of the two source programs from source 25, for every initial register vector. All loop inductions and finite trace constructions are available in PA.

Review evidence. The source checker accepts the exact literal maps and records their hashes in `symmetric_source_certificate.json`. An independent finite regression checks (SG1) on 4,225 guard pairs and (SG2), together with full output equality, on 2,197 triples (t, h, x). Its receipt is `symmetric_duration_regression.json`. These finite checks catch binding or arithmetic transcription errors; the universal justification is the explicit induction above. No new proof-assistant elaboration is claimed by this review.

D Every incoming archive

These entries record supplied contributions and their evidence boundaries. The exact hashes, file counts, byte sizes and provenance are in the machine-readable ledger. A rejected or bounded-search entry is not a lower bound on the state count of a correct machine.

Archive	Contribution and evidence boundary
`Below_U208_Research_2026-10-29.zip`	Full-frame windows, offset quotients and guarded pruning; source 3,909→3,904; no accepted below-208 table.
`ConPA_below208_research_2026-10-29.zip`	Finite-model role-collapse objection, 279,936 fixed-association parameter cases, fold synthesis; no sub-208.
`PA195_ConPA_research_2026-11-14.zip`	Positive195S and195C: unchanged separate190 AX word, 67−6 readers; full source/runtime/startup certificates plus independent literal-tape audits.
`PA_consistency_TM_research_2026-11-13.zip`	159-node five-zero source and 157-node combined representation; exact source CFG checks, no smaller physical allocation.
`U208_ConPA_2026-10-29.zip`	Particular native172/211 history fold reduces a U209 parent to U208.
`U208_ConPA_Conditional_Context_2026-10-29.zip`	Conditional-entry proof; same two original U208 graph classes.
`U208_ConPA_Entry_Context_2026-10-29.zip`	Full entry-context proof; same two original classes.
`U208_ConPA_History_2026-10-29.zip`	Finite-history proof with four exceptional blocks; same original saving.
`conpa_tm_comparison_and_reduction_search.zip`	Experimental MP-result-in-H and MP-without-zero-push 196 variants; rejected Shift layouts halt at 43,563.
`conpa_tm_comparison_and_search.zip`	A/D register-swap 196; rejected two-rotation 195 fails 93 contracts.
`conpa_tm_study_2026-11-13.zip`	Combined196 = H-only Shift + MP-result-in-H: 3,788 sites, 4,928 contracts, 57,922 startup; direct-negation 197 too.
`pa194_conpa_2026-11-13(2).zip`	Different separate190 association: 68 readers and 8 simultaneous folds yield a third positive194 graph. Also contains a distinct intermediate195 separate190 layout with 67 readers and6 folds. The actual intermediate source is3673 sites/4780 contracts despite a clerical report-table diagonal/3700 label.
`pa194_conpa_2026-11-13.zip`	Three-way saving composition: separate190 emitter, 67 raw AX readers and 7 simultaneous folds yield primary PA194; the A/D register permutation gives a second nonisomorphic194. Both include complete 4780 symbolic contracts, exact startup, and a written PA calibration.
`pa195_conpa_2026-11-13.zip`	Positive original PA195: unchanged diagonal191 program, 68−7 reader allocation, detailed ordinary PA calibration.
`pa195_diagonal_combined_2026-11-13.zip`	PA195DC combines bundled diagonal191 and7 folds with cached-head Shift/MP; 3676 sites,4784 contracts,57922 startup, six changed native rows. Optional PA195R and PA195N implement parameter-permuted and negative-vacuity emitters at195, with explicitly limited endpoint claims.
`pa195_followup_research.zip`	M195S8 keeps diagonal191 but pays one extra scanner control and obtains8 folds:135+68-8=195. It rejects the associated194 quotient at source ordinal1003 because seven address zeros trigger the old scanner. Includes additional equivalent source-calculus families and bounded searches.
`pa196_research_2026-11-13.zip`	Exact 157-node representation of 3,812-site original program; loses contextual sharing, no physical reduction.
`pa208_research_2026-10-29.zip`	Eighteen further helper layouts; 27 designated U208 presentations give 20 graphs, not two.

Archive	Contribution and evidence boundary
pa_below196_research.zip	Eight equivalent gates, membership/equality orientations, broad grammar searches; original201-token best 196.
pa_consistency_comparison_2026_11_13.zip	Modal-head/explosion split, twelve semantic compositions, joint gate; physical results 196–200.
pa_consistency_comparison_and_new196_2026-11-13.zip	Diagonal196 (3,700 sites), Separate196 (3,673), Scanner-best196; principal logical/source parents of positive195.
pa_consistency_cross_archive.zip	Four new196 combinations: diagonal/separate AX + Combined controller, each optionally zero-jump; sites 3,676/3,649 and startups57,922/58,168.
pa_consistency_reduction.zip	Rebracketed196, H-only196, integrated gate 203/206; two failed 195 diagnostics.
pa_consistency_reduction_audit_2026-11-14.zip	Four further experimental MP-H compositions, two196 and two197; parameter/order/association/helper/scanner negative searches.
pa_consistency_research.zip	Two exact158-node sources and repaired196; nominal195 needs added scanner state.
pa_consistency_research_2026-10-29.zip	Inductive 3,909→3,904 specialization; no physical sub208.
pa_consistency_research_2026-11-13(2).zip	Twelve-rotation196, diagonal198, successful-MP halting-source experiment; no sub196.
pa_consistency_research_2026-11-13.zip	Earlier diagonal-vacuity199 and short-negation197.
pa_consistency_research_2026-11-14.zip	NEG18: ‘INC7; INC7; node64‘, 18-site encoded negation; complete source3,441 and197 states (134+68−5); independent macro and scratch-zero proofs.
pa_tm_comparison_and_reductions.zip	Two rotated196 graphs; distributed-gate198 with 3,773 source sites.
pa_tm_extended_research_2026-11-13.zip	Four closing-decrement variants and stronger zero invariants; all change the same row, hence not stackable.
pa_tm_research_2026-11-13(2).zip	Zerojump196 (4,959 contracts, 58,402 startup); parity/disequality experiments; rejected195 fails source26.
pa_tm_research_2026-11-13.zip	Shorter-path196 and independent physical checking; detailed direct PA reverse-translation strategy.
pa_unary_power_cover.zip	Unary weak-covering theory and PA proof translations; no new compiled small table.
u208_comparison_research_2026-10-29.zip	Original two-class audit; weak/exact schemas; rejected207.
u208_research_2026-10-29.zip	Primary/alternative U208 and symbolic contracts; same initial history fold.

References

[1] ProveIt, PA consistency-machine overview, pinned commit `768c2be1a8e4d2fcf8aa3443c7b0f084d145734c`. https://github.com/VladimirReshetnikov/ProveIt/blob/768c2be1a8e4d2fcf8aa3443c7b0f084d145734c/Computability/BusyBeaver/Research/PA/README.md.

[2] ProveIt, PA194 bounded halting and conditional consistency endpoint. https://github.com/VladimirReshetnikov/ProveIt/blob/768c2be1a8e4d2fcf8aa3443c7b0f084d145734c/Computability/BusyBeaver/Research/PA/pa194/ipra/README.md.

[3] Supplied primary PA194 packet, `docs/incoming/pa194_conpa_2026-11-13.zip`, especially the construction, independent verifier and report. Its two tables have dense SHA-256 prefixes `4bfea62b17aa` and `27613aa92da7`.

[4] Supplied second PA194 packet, `docs/incoming/pa194_conpa_2026-11-13(2).zip`; eight-fold table with dense SHA-256 prefix `9087d70fc0e1`.

[5] Supplied original PA195 packet, `docs/incoming/pa195_conpa_2026-11-13.zip`, especially the ordinary PA-calibration report and contextual fold proof.

[6] Supplied separate/cached PA195 packet, `docs/incoming/PA195_ConPA_research_2026-11-14.zip`; 195S and 195C construction, command audit and physical verification.

[7] Supplied diagonal/cached PA195 packet, `docs/incoming/pa195_diagonal_combined_2026-11-13.zip`.

[8] Supplied PA195 follow-up, `docs/incoming/pa195_followup_research.zip`; M195S8 and the rejected 194-state scanner quotient.

[9] All archive references above are at the pinned incoming tree: https://github.com/VladimirReshetnikov/ProveIt/tree/768c2be1a8e4d2fcf8aa3443c7b0f084d145734c/docs/incoming.

[10] ProveIt, Cache-return composition on the 217-state source. https://github.com/VladimirReshetnikov/ProveIt/blob/768c2be1a8e4d2fcf8aa3443c7b0f084d145734c/Computability/BusyBeaver/Research/ZFC/candidate-k217/research/cache-return/README.md.

[11] ProveIt, Smaller machines for Con(ZFC), pinned overview. https://github.com/VladimirReshetnikov/ProveIt/blob/768c2be1a8e4d2fcf8aa3443c7b0f084d145734c/Computability/BusyBeaver/Research/ZFC/README.md.

[12] ProveIt, Concise presentations of intuitionistic primitive recursive arithmetic. https://github.com/VladimirReshetnikov/ProveIt/blob/768c2be1a8e4d2fcf8aa3443c7b0f084d145734c/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA/README.md.